VICTORY 2027

The Excretory System



LECTURE 3 | BIOLOGY

Upasana Ma'am



Topics To Be Covered

1. URINE FORMATION
2.

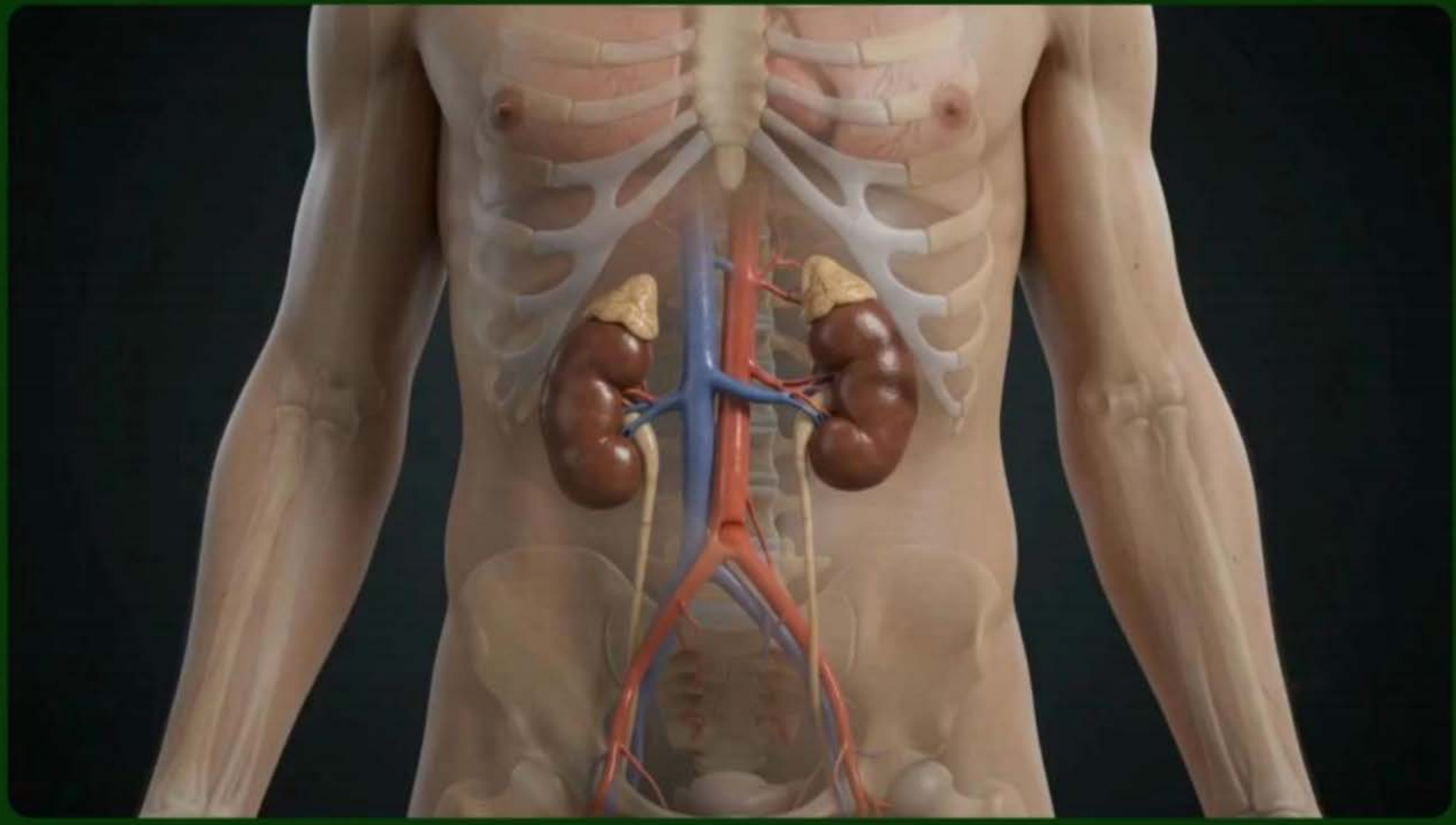
RATTON KO KARO **BYE BYE !!**

EASY CONCEPTS SE KARO **HELLO HI !!**

With – *Uppu Ma'am*



KIDNEY



KIDNEY



QUESTIONS

Loop of Henle lies in:

(ICSE 2021-22 Sem II)

A Medulla

B Cortex

C Pelvis

D Ureter

QUESTIONS

The nephron discharge their urine at the:

(ICSE 2016)

A Urinary bladder

B Urethra

C Renal pelvis

D Renal Pyramid

QUESTIONS

The pale yellow colour of normal human urine is due to the pigment:
(ICSE 2021-22 Sem II)

A Melanin

B Anthocyanin

C Urochrome

D Haemoglobin

QUESTIONS

Identify the structure that helps in the removal of nitrogenous waste as well as concerned with osmoregulation.

A Protonephredia

B Malphigian tubules

C Nephredia

D All of the above

QUESTIONS

The main function of Distal convoluted Tubule (DCT) of nephron is to maintain the:

A pH in blood

B Na-K balance of blood

C Temperature of blood

D Both (1) and (2)

QUESTIONS

The animals called ammoniotelic excretes mainly:

A. Urea

B. Uric acid

C. Ammonia

D. Both (1) and (2)

QUESTIONS

The ultrafiltration takes place in:

A Glomerulus

B PCT

C DCT

D Collecting duct

QUESTIONS

The outer darker region is called:

A Cortex

B Medulla

C Nephron

D Bowman's capsule

QUESTIONS

Filtration of any liquid under tremendous pressure is called:

A Ultrafiltration

B Selective reabsorption

C Tubular secretion

D Both (1) and (2)

QUESTIONS

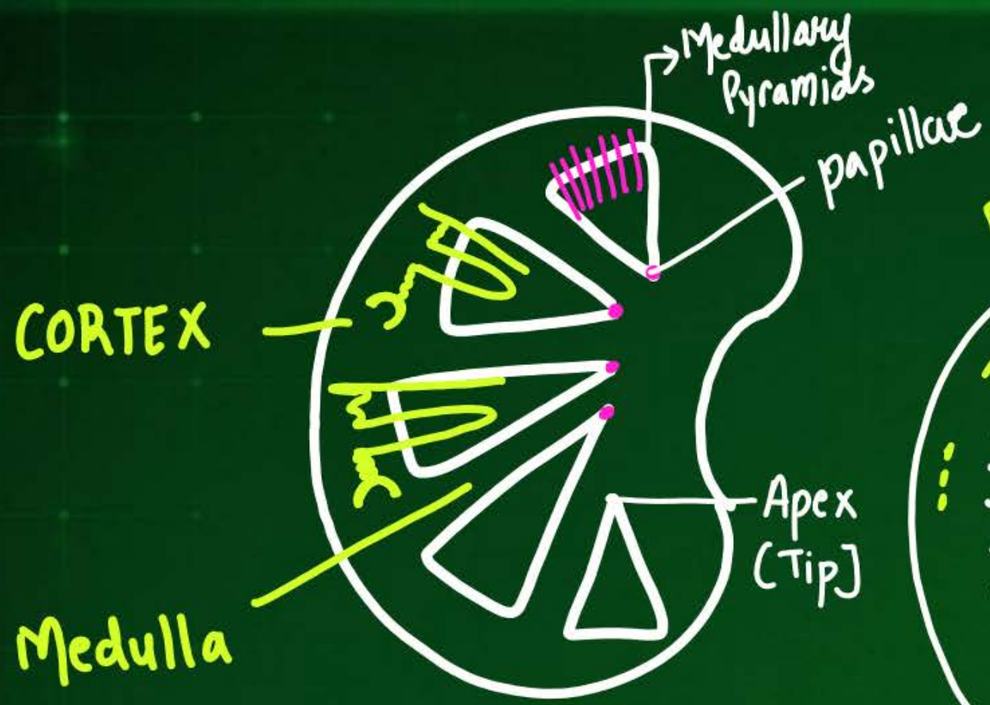
The opening of bladder into the urethra is guarded by a:

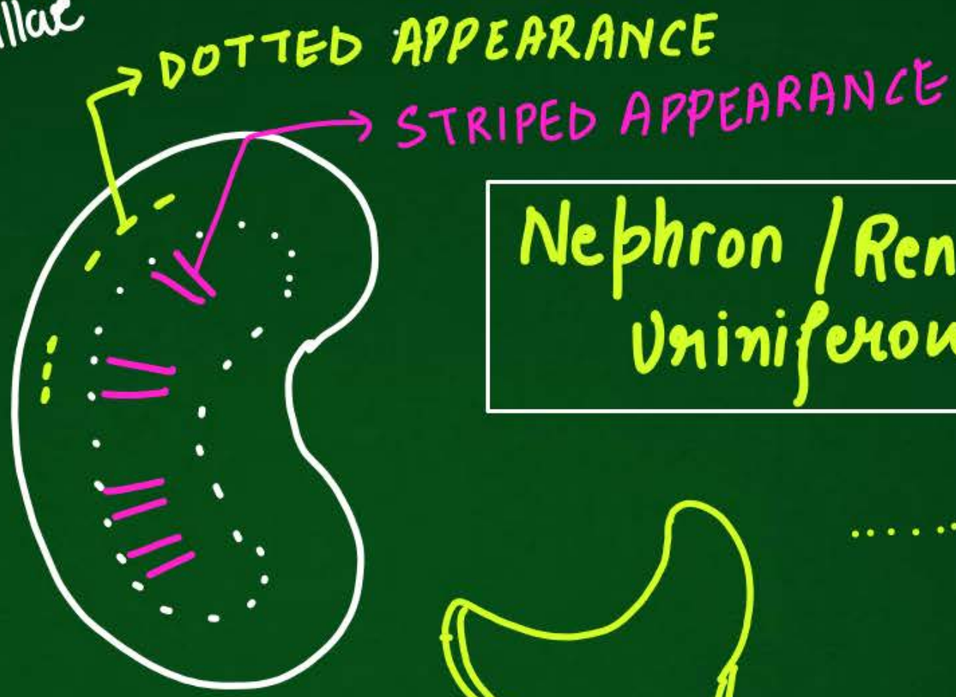
A Sphincter muscle

B Pelvis

C Vasa recta

D Both (1) and (2)





Nephron / Renal Tubule / Uniniferous Tubule

Dothead

CORTEX

→ Bowman's Capsule

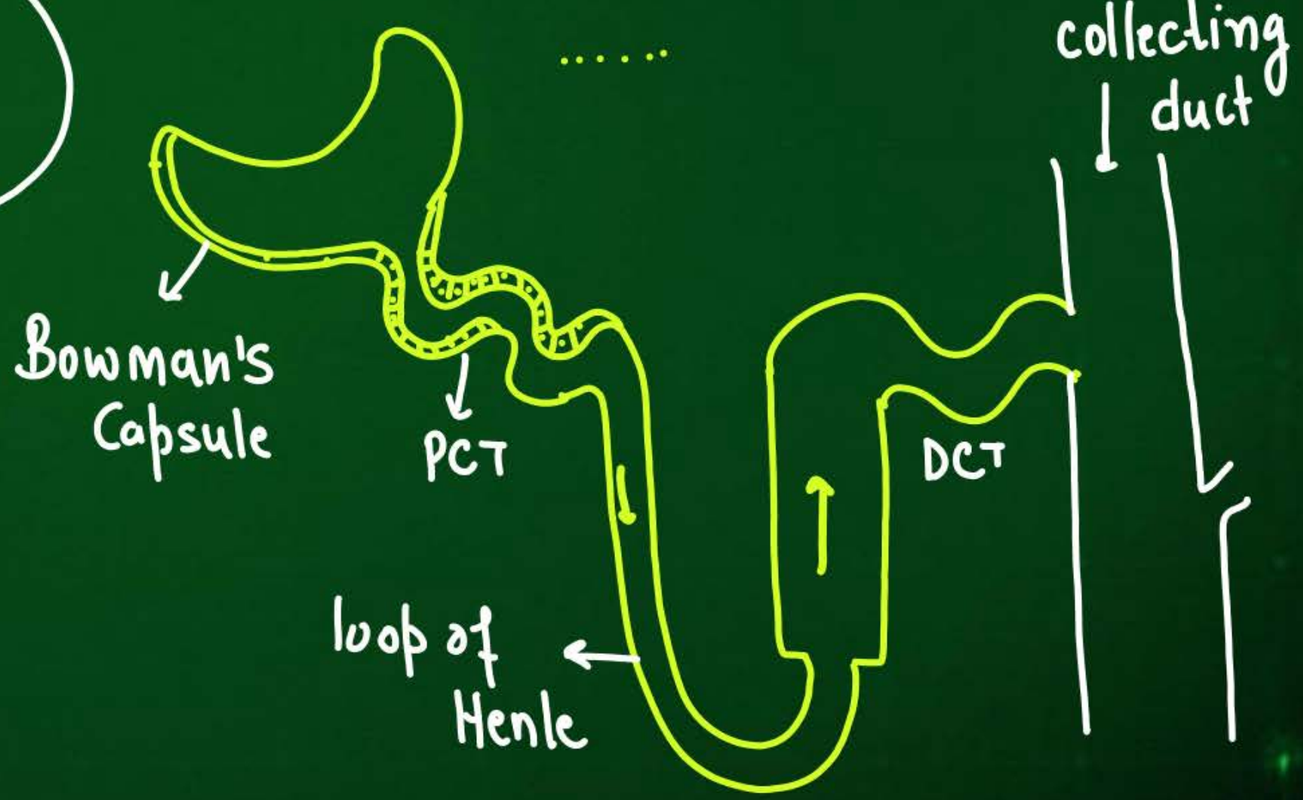
→ PCT, DCT

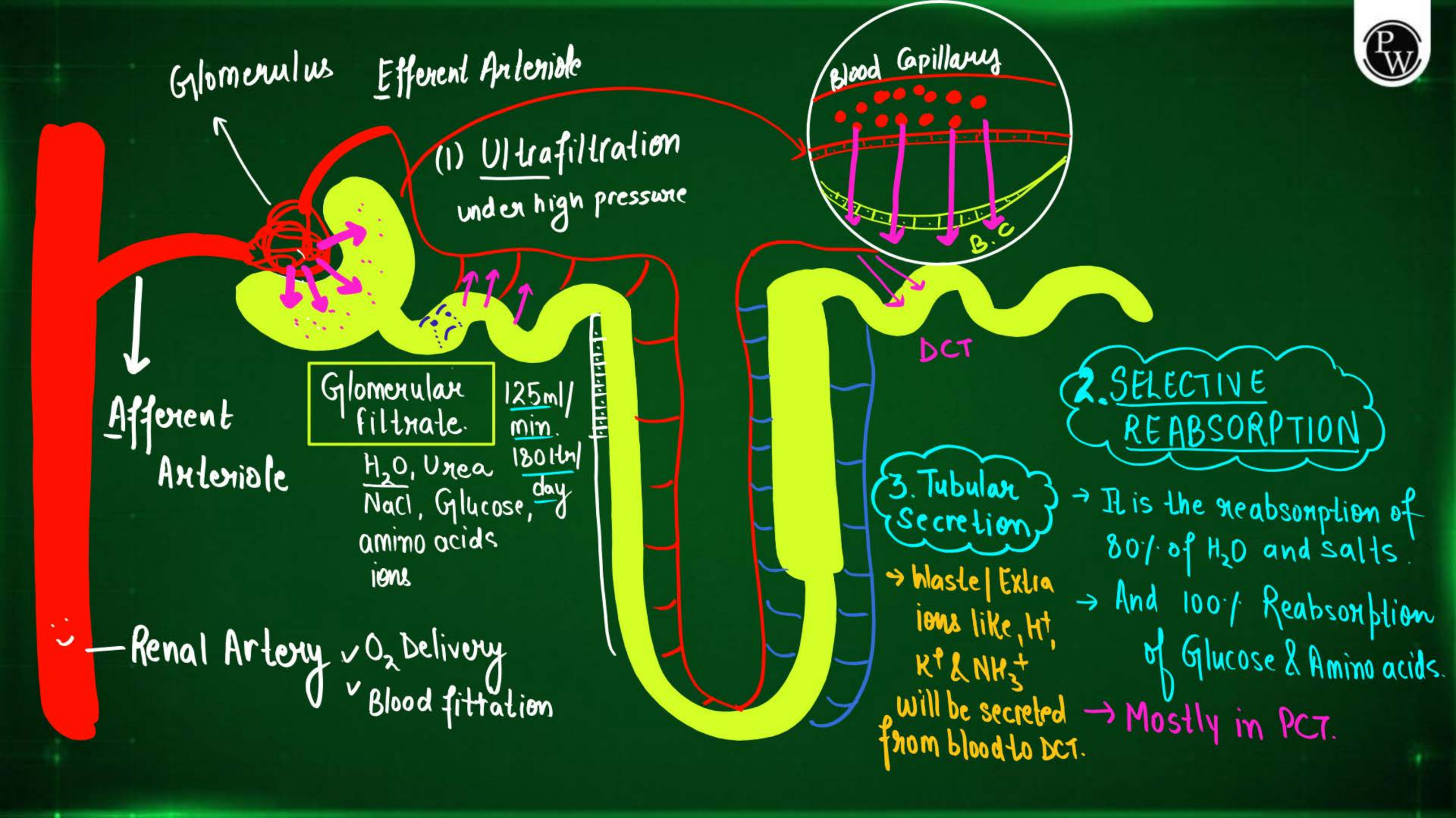
→ [Half] LOH & Collecting Duct [CD]

Striped

MEDULLA

→ Half LOH & CD




Glomerulus
Efferent Arteriole
Blood Capillary
(1) Ultrafiltration
under high pressure
B.C
Afferent Arteriole
Glomerular filtrate.
125ml/min.
180ltr/day
H_2O, Urea
NaCl, Glucose,
amino acids
ions
DCT
Renal Artery
↓ O_2 Delivery
↓ Blood filtration
2. SELECTIVE REABSORPTION
3. Tubular Secretion
→ It is the reabsorption of 80% of H_2O and salts.
→ And 100% Reabsorption of Glucose & Amino acids.
→ Waste / Extra ions like, H^+, K^+ & NH_3^+ will be secreted from blood to DCT.
→ Mostly in PCT.

Thank You